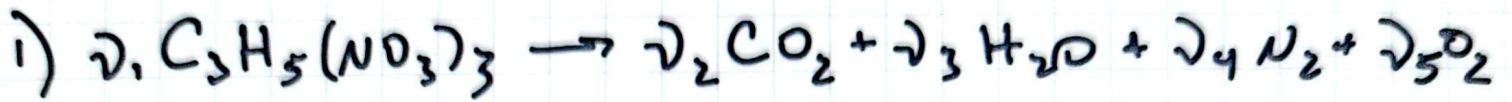


# Homework 1

Nanya Mijamethans



$$C: 3\nu_1 = 1\nu_2 + 0\nu_3 + 0\nu_4 + 0\nu_5$$

$$O: 9\nu_1 = 2\nu_2 + 1\nu_3 + 0\nu_4 + 2\nu_5$$

$$N: 3\nu_1 = 0\nu_2 + 0\nu_3 + 2\nu_4 + 0\nu_5$$

$$H: 5\nu_1 = 0\nu_2 + 2\nu_3 + 0\nu_4 + 0\nu_5$$

Basis:  $\nu_1 = 1$  ✓

$$C: 3 = 1\nu_2 + 0\nu_3 + 0\nu_4 + 0\nu_5$$

$$O: 9 = 2\nu_2 + 1\nu_3 + 0\nu_4 + 2\nu_5$$

$$N: 3 = 0\nu_2 + 0\nu_3 + 2\nu_4 + 0\nu_5$$

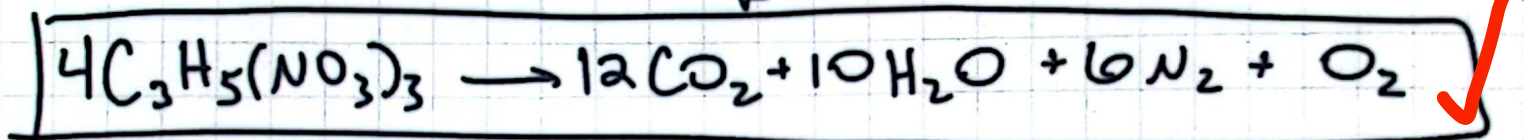
$$H: 5 = 0\nu_2 + 2\nu_3 + 0\nu_4 + 0\nu_5$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 2 \\ 0 & 0 & 2 & 0 \\ 0 & 2 & 0 & 0 \end{bmatrix}^A \times \begin{bmatrix} \nu_2 \\ \nu_3 \\ \nu_4 \\ \nu_5 \end{bmatrix}^B = \begin{bmatrix} 3 \\ 9 \\ 3 \\ 5 \end{bmatrix}^C$$

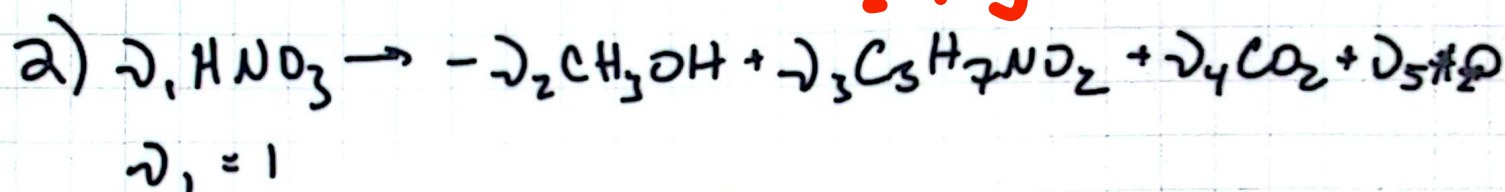
$$B = A^{-1} \times C$$

$$\nu_1 = 1 \quad \nu_2 = 3 \quad \nu_3 = 2.5 \quad \nu_4 = 1.5 \quad \nu_5 = 0.25$$

↓ × 4

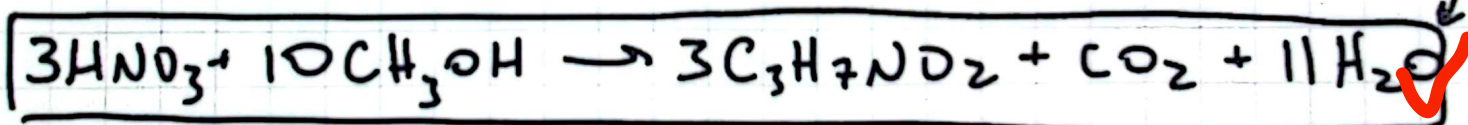


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$$\begin{matrix} C \\ H \\ O \\ N \end{matrix} \begin{bmatrix} 0 \\ 1 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 & -3 & -1 & 0 \\ 4 & 7 & 0 & 2 \\ 1 & 2 & 2 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} \nu_2 \\ \nu_3 \\ \nu_4 \\ \nu_5 \end{bmatrix}$$

$$\nu_1 = 1 \quad \nu_2 = -3.3 \quad \nu_3 = 1 \quad \nu_4 = 0.3 \quad \nu_5 = 3.6 \quad \checkmark$$



$$\frac{44 \text{ mg } HNO_3}{1 \text{ L } H_2O} \times \frac{1 \text{ g}}{1000 \text{ mg}} \times \frac{1 \text{ mol } HNO_3}{63.01 \text{ g } HNO_3} \times \frac{10 \text{ mol } MeOH}{3 \text{ mol } HNO_3} \times \frac{32.04 \text{ g } MeOH}{1 \text{ mol } MeOH}$$

$$= 0.0746 \text{ g/L MeOH}$$

$$\times 10 \text{ L}$$

$$0.746 \text{ g methanol} \quad \checkmark$$

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5)  $V$  = volume of tank  $L$

$$\left(A\left(\frac{L}{h}\right) + B\left(\frac{L}{h}\right)\right) \cdot 0.5(h) = V(L)$$

$$\begin{aligned} 0.5(A+B) &= V \\ 0.75(A+C) &= V \\ B+C &= V \end{aligned} \quad \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{3}{4} & 0 & \frac{3}{4} \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Basis:  $V=1$

$$A = 1.67 \frac{L}{h} \quad B = 0.83 \frac{L}{h} \quad C = 0.17 \frac{L}{h}$$

a)  $t(1.67 + 0.83 + 0.17) = 1$

$t = 0.462 \text{ hours}$  ✓

b) A:  $1.67 t = 1$   $t_A = 0.857 \text{ hours}$  ✓

B:  $0.83 t = 1$   $t_B = 1.200 \text{ hours}$  ✓

C:  $0.17 t = 1$   $t_C = 6 \text{ hours}$  ✓

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$$\begin{aligned}
 3) \text{ C}_2\text{H}_5\text{OH} &: \frac{6(1.01) \text{ g/mol}}{46.068 \text{ g/mol}} = 13.15 \text{ wt \% H} \\
 \text{CH}_4 &: \frac{4(1.01) \text{ g/mol}}{16.04 \text{ g/mol}} = 25.19 \text{ wt \% H} \\
 \text{H}_2\text{O} &: \frac{2(1.01) \text{ g/mol}}{18.015 \text{ g/mol}} = 11.21 \text{ wt \% H} \\
 \text{C}_6\text{H}_{12}\text{O}_6 &: \frac{12(1.01) \text{ g/mol}}{180.156 \text{ g/mol}} = 66.75 \text{ \% H}
 \end{aligned}$$

No comments, -5 pts

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4) see Excel File

Should have explained comparisons between data, -10 pts

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